

When Can a Machine Judge Like a Human?

A PRISMA Systematic Review of LLM-as-a-Judge versus Human Annotation across 29 Domains and Tasks

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Abstract

Large language models (LLMs) are increasingly deployed as automatic *judges* and *annotators*. Adoption has outpaced validation, and the field lacks a panoramic answer to the question that matters: *not* “do LLM judges work?” but “*in which domains, tasks, and conditions* do LLM judgments agree with human judgments well enough to substitute for them?” We address this with a PRISMA 2020 systematic review. Using a triangulated identification strategy—OpenAlex (1500 records, 957 unique), Semantic Scholar, and a web deep-search—we screened 967+ records and included 217 studies across 29 domains. Each was coded on a five-level reliability rubric benchmarked against the *human-human* ceiling. Only 31% of studies place LLM judges at or near validated substitution; the modal verdict is *mixed/task-dependent* (mean reliability 3.15/5). Reliability is high—often exceeding the human ceiling—for objective, well-structured, low-expertise tasks judged at the aggregate level, and collapses for subjective, expert, low-resource, or adversarial settings. We provide a domain×task reliability landscape, a deployment decision map, a catalogue of failure modes, and a validation checklist.

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1 Introduction

The cost, latency, and irreproducibility of human evaluation have made “LLM-as-a-judge” the dominant scalable surrogate [13, 37]. Yet a model that matches human preferences at >80% on chat [37] can correlate near zero with expert creativity judgments [4], or be fooled into mislabeling relevance by a few injected keywords [1]. This review maps, domain by domain and task by task, the conditions under which LLM–human agreement is good enough to substitute, anchoring every claim to the human–human ceiling. Contributions: a PRISMA review of 217 studies across 29 domains; a reproducible five-level reliability rubric; a reliability landscape and decision map; and synthesis along the judge-architecture axis.

2 Background and related reviews

LLM-as-a-judge descends from reference-free metrics but uses a general instruction-following model with a natural-language rubric [12, 22]. Four architectural families recur: prompted general LLMs [26, 37]; fine-tuned open evaluators [19, 21, 33, 34, 38]; juries/multi-agent panels [5, 32]; and calibrated rubric methods [8, 16]. Prior surveys taxonomize methodology [13, 23]; large meta-evaluations supply task-level data [2, 3, 15, 31]. This review synthesizes the cross-domain reliability map they imply.

3 Methods

We followed PRISMA 2020 [27]. **Inclusion:** empirical studies (2023–2026) that directly compare LLM judgments to human judgments and report an agreement metric or structured concordance. Identification triangulated OpenAlex (primary; 1500 retrieved, 957 unique), Semantic Scholar, and a 59-query web search (Figure 1). Each study was coded for domain, task formulation, subjectivity, expertise, judge model/architecture, comparator, metric, headline agreement, human baseline, biases, and a five-level reliability verdict (1 Validated ... 5 Unreliable) benchmarked against the human ceiling; reliability score = 6–verdict.

4 Results

4.1 Corpus overview

The 217 studies span 29 domains (Figure 2a), concentrated in methodology/bias, general NLG, education, medicine, and social-science annotation; by venue the corpus is ML/NLP-dominant with HCI/CSCW and domain-journal representation. Volume rises across 2023–2026 (Figure 2b); the objective/mixed/ subjective split (Figure 3) is the strongest predictor of reliability.

4.2 Overall and by domain

The modal verdict is mixed; 31% reliable, 19% problematic (mean 3.15/5; Figure 4). **Green (≥ 3.5 ; 8 domains):** MT (system level), text-to-image, instruction-following, QA/factuality, information extraction, general-NLG pairwise, dialogue, legal. **Amber (19):** summarization, RAG, education, social-science annotation, medicine (structure-dependent), code, IR/relevance (contested [6, 10]).

Figure 1. PRISMA 2020 flow — two-arm identification (OpenAlex database + web/citation)

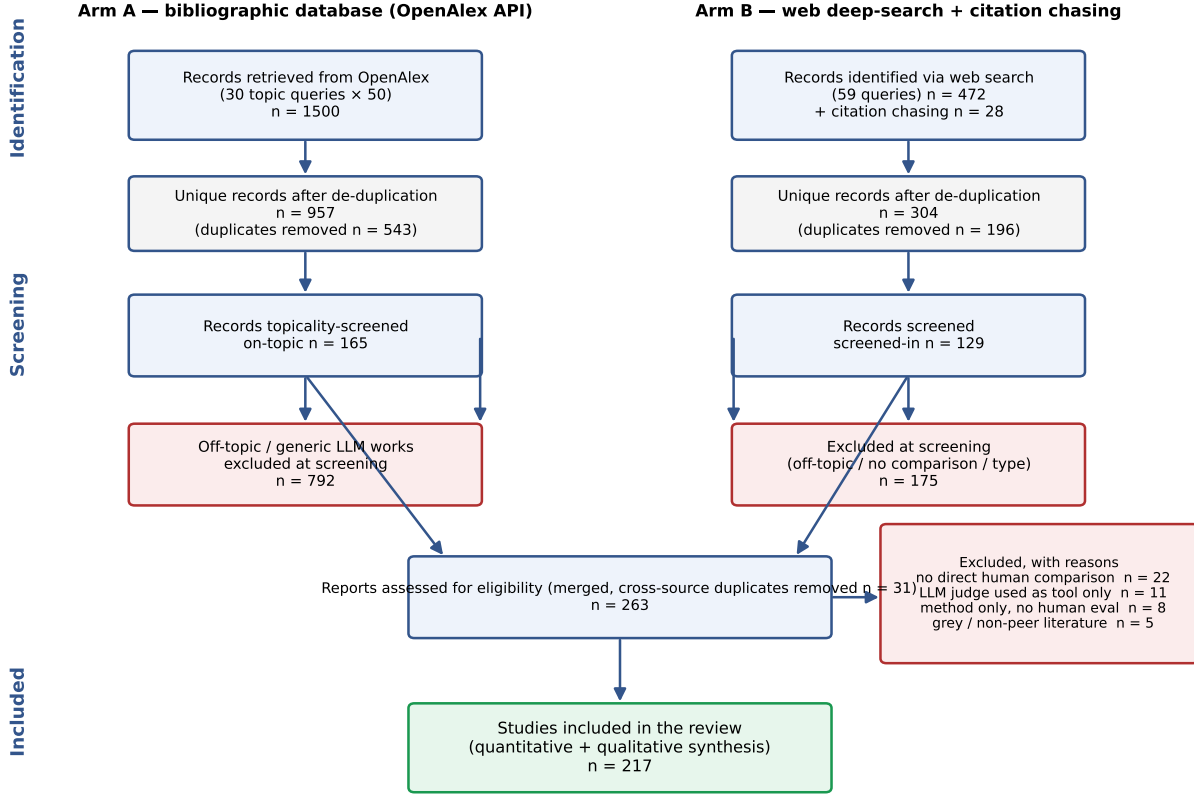


Figure 1: PRISMA 2020 two-arm identification (OpenAlex database + web/citation).

Red (2): low-resource multilingual, creative writing. Family aggregates appear in Figure 6; the domain and landscape views in Figures 5a–5b; the decision table in Table 1.

4.3 Task type, subjectivity, expertise

Verification/classification tasks are more reliable than open-ended scoring (Figure 7c); construct subjectivity and required expertise both predict reliability monotonically (Figure 7a,b), and judges align better with non-experts than experts [2, 30].

4.4 Agreement vs. the human ceiling

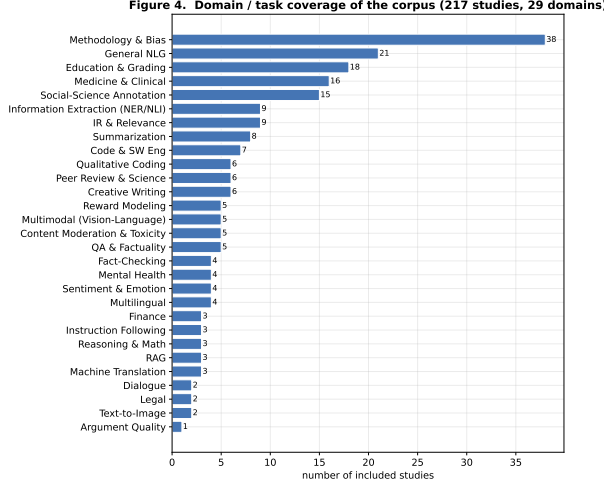
The 29 studies with numeric values (Figure 8) show the strongest results meet or exceed the human ceiling—MT-Bench 85% vs. 81% [37], GEMBA-MQM 96.5% [20], SAFE wins 76% of disagreements [36], Prometheus Pearson 0.897 [19], PRISMA screening $\kappa > 0.9$ [14]—while the weakest fall far below (JUDGE-BENCH $\kappa = 0.28$ [2]). On subjective tasks the human ceiling is itself low ($\alpha < 0.35$ [18]; Figure 9; Table 2).

Table 1: Domain-level decision table. Mean reliability score (1 = unreliable ... 5 = validated), modal verdict, and traffic-light signal (GREEN ≥ 3.5 ; AMBER 2.5–3.5; RED < 2.5). Sorted by mean reliability. Small- n domains should be read together with the family-level aggregates.

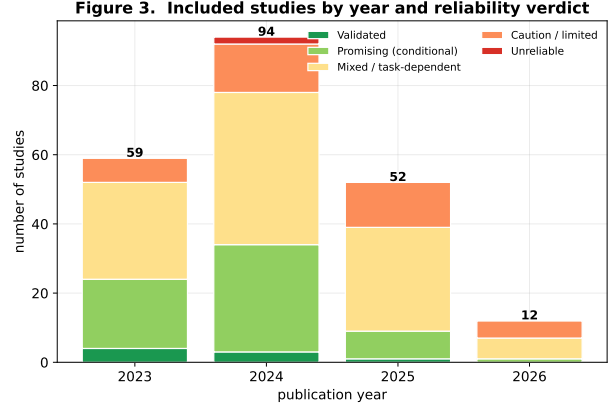
Domain / task	n	Mean rel.	Modal verdict	Signal
Machine Translation	3	4.33	Promising	GREEN
Text-to-Image	2	4.00	Promising	GREEN
Instruction Following	3	4.00	Validated	GREEN
QA & Factuality	5	3.80	Promising	GREEN
Information Extraction (NER/NLI)	9	3.78	Promising	GREEN
General NLG	21	3.71	Promising	GREEN
Dialogue	2	3.50	Promising	GREEN
Legal	2	3.50	Promising	GREEN
Education & Grading	18	3.39	Mixed	AMBER
RAG	3	3.33	Mixed	AMBER
Summarization	8	3.25	Promising	AMBER
Fact-Checking	4	3.25	Promising	AMBER
Sentiment & Emotion	4	3.25	Mixed	AMBER
IR & Relevance	9	3.22	Promising	AMBER
Social-Science Annotation	15	3.20	Mixed	AMBER
Peer Review & Science	6	3.17	Mixed	AMBER
Multimodal (Vision-Language)	5	3.00	Mixed	AMBER
Qualitative Coding	6	3.00	Mixed	AMBER
Argument Quality	1	3.00	Mixed	AMBER
Finance	3	3.00	Mixed	AMBER
Medicine & Clinical	16	2.88	Mixed	AMBER
Methodology & Bias	38	2.87	Mixed	AMBER
Reward Modeling	5	2.80	Mixed	AMBER
Code & SW Eng	7	2.71	Mixed	AMBER
Reasoning & Math	3	2.67	Mixed	AMBER
Content Moderation & Toxicity	5	2.60	Mixed	AMBER
Mental Health	4	2.50	Mixed	AMBER
Creative Writing	6	2.33	Mixed	RED
Multilingual	4	2.25	Caution	RED

Table 2: Quantitative anchors: studies reporting an extractable numeric LLM–human agreement value, with the human–human baseline where reported. Metrics are heterogeneous (see §3) and are interpreted within metric families.

Study	Domain	Metric	LLM–human	Human–human	Verdict
Chakrabarty et al. (2024)	Creative Writing	pearson	0.10		Unreliable
Fein et al. (2025)	Creative Writing	pct_agreement	0.78		Mixed
Lee et al. (2024)	Education & Grading	qwk	0.57	0.66	Mixed
Henkel et al. (2024)	Education & Grading	kappa	0.70	0.75	Promising
Mizumoto & Eguchi (2023)	Education & Grading	qwk	0.74		Promising
Zheng et al. (2023)	General NLG	pct_agreement	0.85	0.81	Validated
Li et al. (2024)	General NLG	pct_agreement	0.89		Promising
Kim et al. (2023)	General NLG	pearson	0.90		Promising
Dubois et al. (2024)	General NLG	winrate_corr	0.98		Promising
Dubois et al. (2023)	General NLG	winrate_corr	0.98		Promising
Lin et al. (2024)	General NLG	winrate_corr	0.98		Promising
Pradhan et al. (2025)	Legal	spearman	0.73		Promising
Kocmi & Federmann (2023)	Machine Translation	pct_agreement	0.96		Validated
Szymanski et al. (2025)	Medicine & Clinical	pct_agreement	0.66		Caution
Brake et al. (2025)	Medicine & Clinical	icc	0.82		Mixed
Mitsuyama et al. (2024)	Medicine & Clinical	accuracy	0.94		Mixed
Bavaresco et al. (2024)	Methodology & Bias	kappa	0.28		Mixed
Koo et al. (2024)	Methodology & Bias	winrate_corr	0.44		Caution
Yasunaga et al. (2025)	Multimodal (Vision-Language)	accuracy	0.72		Caution
Chen et al. (2024)	Multimodal (Vision-Language)	pct_agreement	0.79		Mixed
Ho et al. (2024)	Peer Review & Science	pct_agreement	0.07		Caution
Tarakji et al. (2024)	Peer Review & Science	pct_agreement	0.61		Mixed
Guo et al. (2024)	Peer Review & Science	kappa	0.91		Validated
Wei et al. (2024)	QA & Factuality	pct_agreement	0.72		Promising
Xiao et al. (2025)	Qualitative Coding	fleiss_kappa	0.46		Mixed
Tan et al. (2024)	Reasoning & Math	accuracy	0.55	0.50	Caution
Wen et al. (2026)	Reward Modeling	kendall	0.61	0.76	Caution
Pangakis et al. (2023)	Social-Science Annotation	f1	0.71		Mixed
Liu et al. (2023)	Summarization	spearman	0.51		Promising



(a)



(b)

Figure 2: Coverage: (a) studies per domain; (b) studies per year by verdict.

Table 3: Validation checklist for trusting an LLM judge on a new task.

Question	If “no”/unfavourable
Objective with verifiable ground truth?	Expect verdict 3–5
Low required expertise; non-expert labels suffice?	Beware expert–LLM gap
Human–human baseline known (“match a second human”)?	Run the alt-test [3]
Right granularity (item vs. system)?	Re-test at item level
Comparative (pairwise/jury) not absolute?	Prefer comparative [32]
Presentation biases controlled?	Debias [7]
Generator and judge unrelated?	Avoid preference leakage [24]
Temperature/seed/version pinned?	Pin [25]

4.5 Failure modes and moderators

Recurring biases—verbosity/length [7], position [37], self-preference [28, 35], style-over-substance [9], preference leakage [24], non-determinism [25], the expertise gap, and adversarial manipulability [29]—modulate reliability (Figure 10). Moderators: fine-tuned judges excel in-domain but do not generalize [17, 38]; juries reduce bias [32]; aggregate $\neq$ item-level reliability [20]; low-resource languages degrade sharply [11].

5 Discussion

Green zone (automate w/ light audit): objective, verifiable, low-expertise tasks judged in aggregate. **Amber zone (human-in-the-loop + per-task validation):** summarization, RAG, grading, social-science annotation, peer review, structured clinical evaluation. **Red zone (keep humans):** subjective/expert/ low-resource/adversarial judgments. *Architecture matters:* comparative > absolute, juries > single, calibrated > raw, fine-tuned = in-domain specialist. The decision map (Figure 11) and checklist (Table 3) operationalize this.

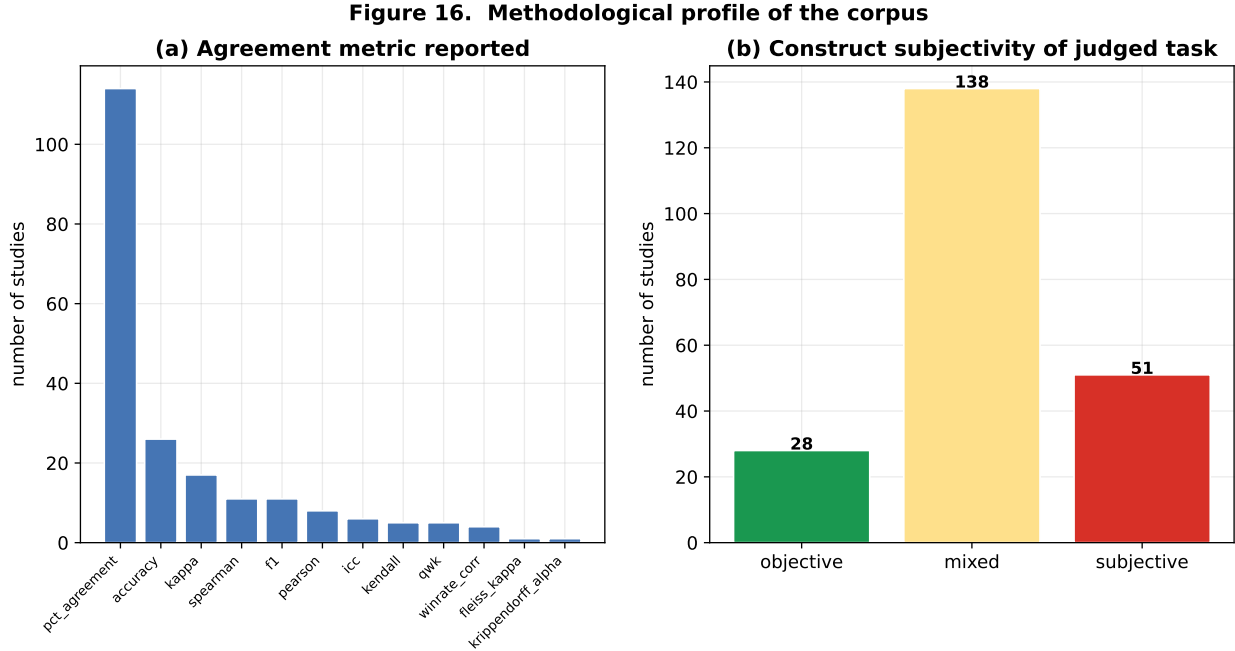


Figure 3: Methodological profile: metric reported and construct subjectivity.

6 Limitations

Single-reviewer LLM-assisted screening/extraction; heterogeneous metrics synthesized within families; small n in several domains; database recall depends on indexing; findings reflect mid-2026 models.

7 Conclusion

LLM-as-a-judge is a *validated* substitute for objective, aggregate-level tasks, a *conditional* human-in-the-loop assistant for many applied domains, and an *unsafe* replacement for subjective/expert/low-resource/adversarial judgments. The right question is “can this judge replace a second human on this task, at this granularity, under these controls?”—answerable domain by domain.

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Figure 2. Reliability verdict across all included studies

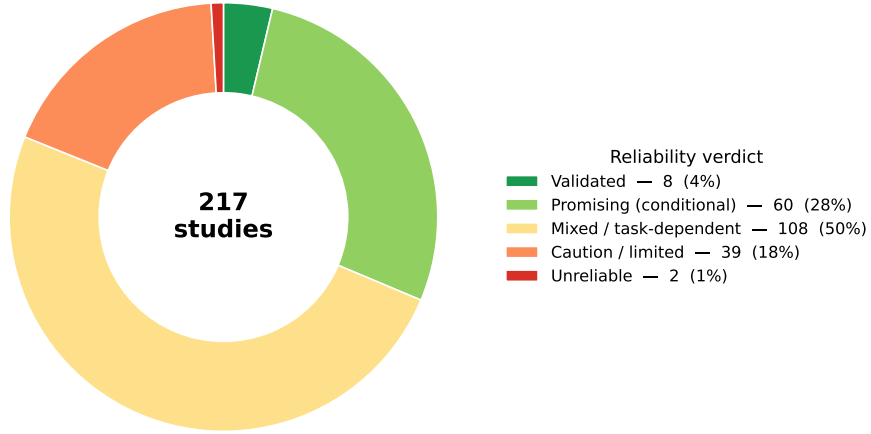
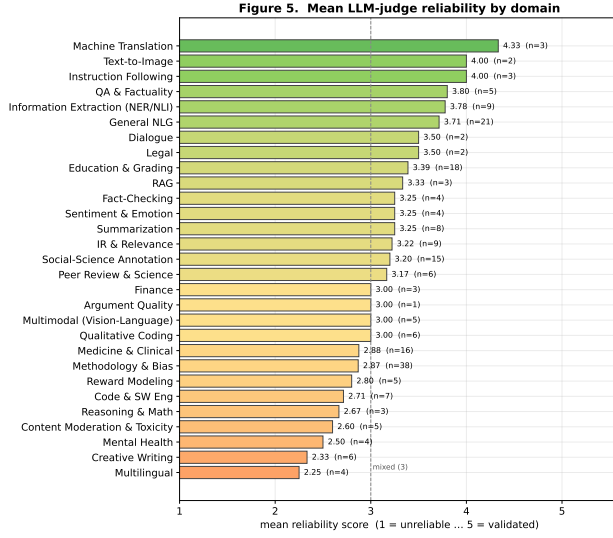
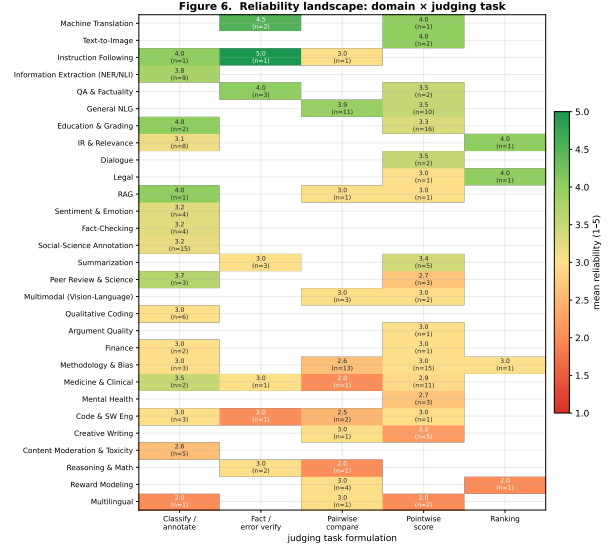


Figure 4: Reliability verdict across all 217 studies.

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(a)



(b)

Figure 5: (a) Mean reliability by domain. (b) Domain×task reliability landscape.

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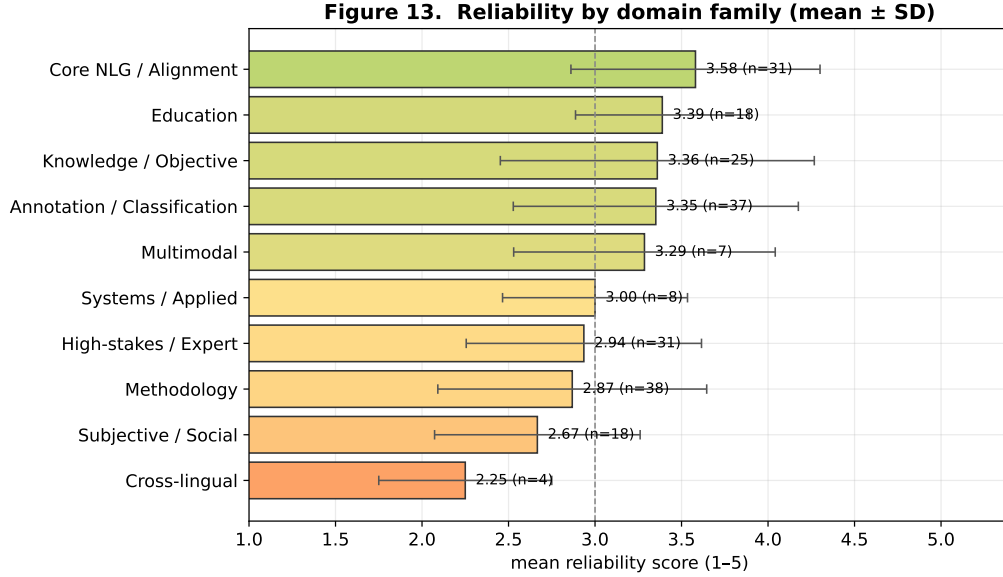


Figure 6: Reliability by domain family (mean $\pm$ SD).

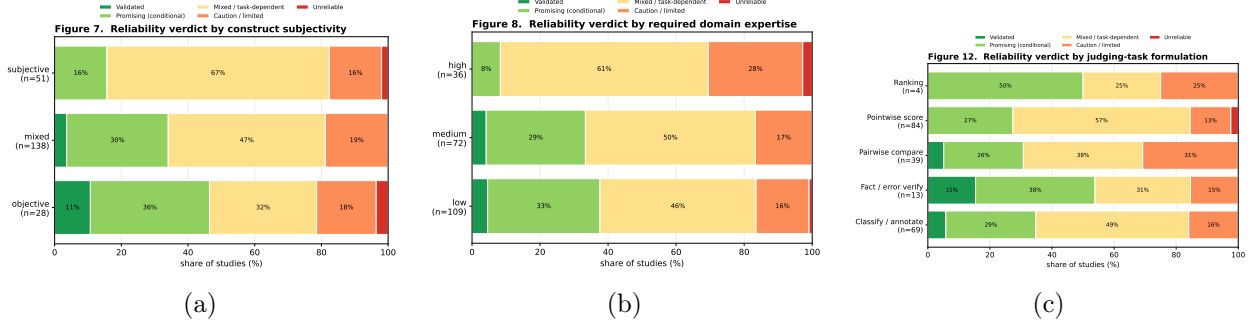


Figure 7: Verdict by (a) subjectivity, (b) expertise, (c) task formulation.

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Figure 9. Reported LLM-human agreement, by study (grouped by metric family)

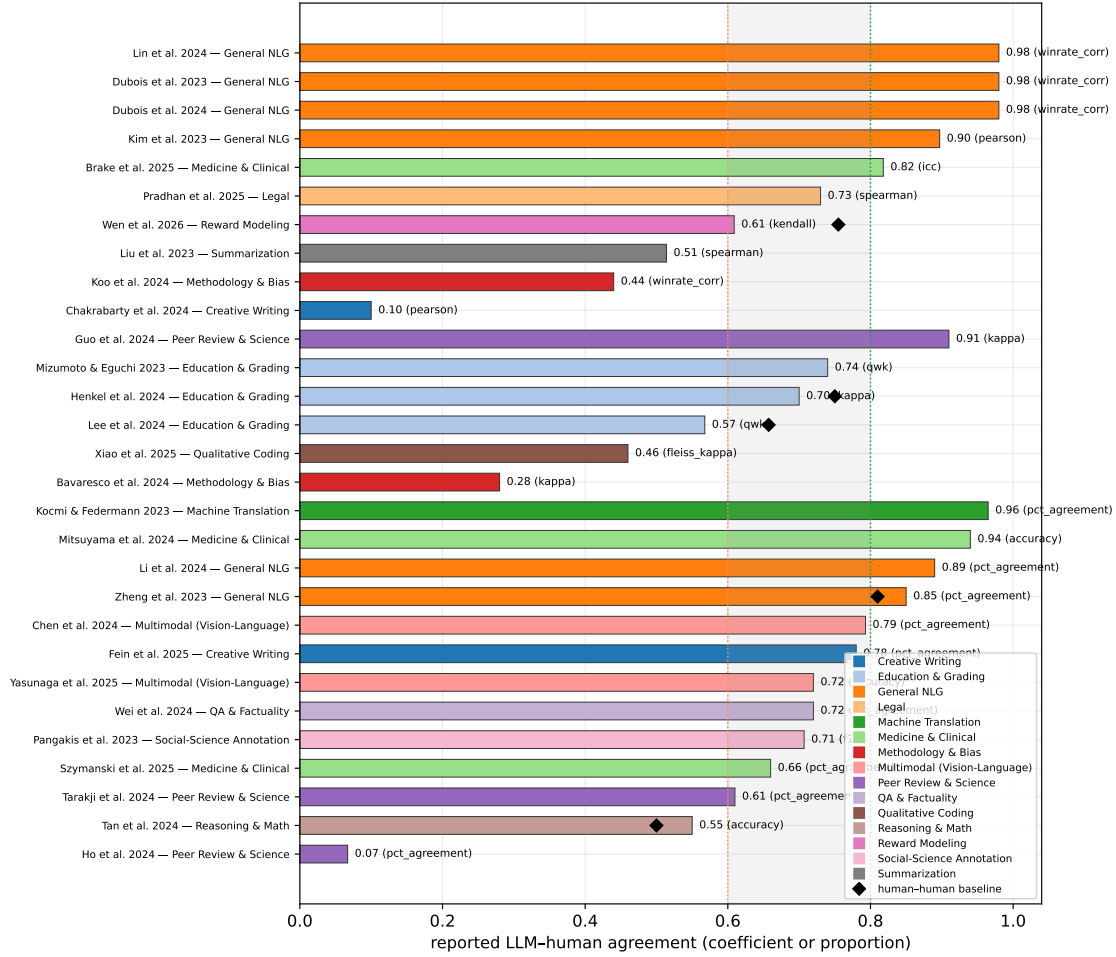


Figure 8: Reported LLM-human agreement by study, grouped by metric family.

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Figure 10. LLM-human vs human-human agreement (studies reporting both)

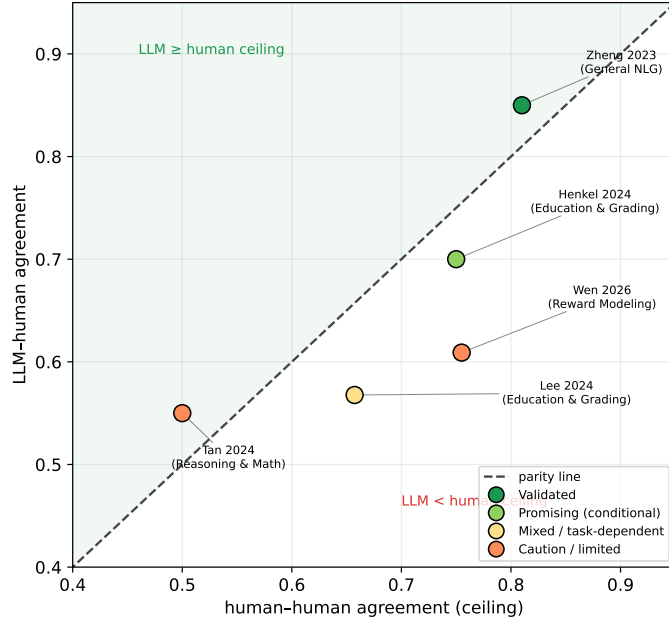


Figure 9: LLM-human vs. human-human agreement.

A Full extracted-studies corpus

Table 4: Full extracted-studies corpus (217 included studies). Agreement is the headline agreement coefficient/proportion where reported.

Key	Study	Domain	Task	Metric
mirzakhmedova2024argument	Mirzakhmedova et al. (2024)	Argument Quality	pointwise	pct_
tong2024codejudge	Tong et al. (2024)	Code & SW Eng	classify	accu
watson2024aicode	Watson et al. (2024)	Code & SW Eng	classify	pct_
codespec2025	Jin et al. (2025)	Code & SW Eng	verify	accu
wang2025setesting	Wang et al. (2025)	Code & SW Eng	pairwise	pct_
zhuo2025codejudgeeff	Wang et al. (2025)	Code & SW Eng	pointwise	kapp
biasloop2026	Zhao et al. (2026)	Code & SW Eng	pairwise	pct_
overcorrect2026	Jin et al. (2026)	Code & SW Eng	classify	accu
hatespeechprobe2023	Roy et al. (2023)	Content Moderation & Toxicity	classify	f1
li2023hotchatgpt	Li et al. (2023)	Content Moderation & Toxicity	classify	pct_
nguyen2024crisismisinfo	Nguyen & Rudra (2024)	Content Moderation & Toxicity	classify	f1
hatespeechanno2025	authors (2025)	Content Moderation & Toxicity	classify	kapp
icwsm2025hatebias	Giorgi et al. (2025)	Content Moderation & Toxicity	classify	kapp
xie2023nextchapter	Xie et al. (2023)	Creative Writing	pointwise	pct_
chakrabarty2024artartifice	Chakrabarty et al. (2024)	Creative Writing	pointwise	pear
ismayilzada2024creative	Ismayilzada et al. (2024)	Creative Writing	pointwise	pct_
creativeevalmdpi2025	Kim et al. (2025)	Creative Writing	pointwise	spea
haase2025creativitypeaked	Haase et al. (2025)	Creative Writing	pointwise	pct_
litbench2025	Fein et al. (2025)	Creative Writing	pairwise	pct_
lin2023llmeval	Lin & Chen (2023)	Dialogue	pointwise	spea
mendonca2024dialogue	Mendonca et al. (2024)	Dialogue	pointwise	spea
hackl2023reliablerater	Hackl et al. (2023)	Education & Grading	pointwise	icc
khademi2023bardassessment	Khademi (2023)	Education & Grading	pointwise	pct_
kortemeyer2023physics	Kortemeyer (2023)	Education & Grading	pointwise	pct_

Key	Study	Domain	Task	Metri
mizumoto2023aes	Mizumoto & Eguchi (2023)	Education & Grading	pointwise	qwk
yancey2023cefr	Yancey et al. (2023)	Education & Grading	pointwise	qwk
aescomparative2024	Lee et al. (2024)	Education & Grading	pointwise	qwk
floden2024gradingexams	Flodén (2024)	Education & Grading	pointwise	pct
grevisse2024asagmed	Grévisse (2024)	Education & Grading	classify	pct
hou2024classroom	Hou et al. (2024)	Education & Grading	pointwise	icc
k12shortanswer2024	Henkel et al. (2024)	Education & Grading	classify	kapp
koraishi2024langassess	Koraishi (2024)	Education & Grading	pointwise	pct
quah2024dentalaes	Quah et al. (2024)	Education & Grading	pointwise	icc
wang2024beyondagreement	Wang et al. (2024)	Education & Grading	pointwise	qwk
writingmultidim2024	Steiss et al. (2024)	Education & Grading	pointwise	qwk
asagfair2025	Henkel et al. (2025)	Education & Grading	pointwise	kapp
sefl2025feedback	authors (2025)	Education & Grading	pointwise	f1
studentsjudge2025	Banihashem et al. (2025)	Education & Grading	pointwise	pct
k12sci2026	He et al. (2026)	Education & Grading	pointwise	pct
claimmatch2023	Choi et al. (2023)	Fact-Checking	classify	pct
ni2024afacta	Ni et al. (2024)	Fact-Checking	classify	pct
politicaltruths2024	Chatrath et al. (2024)	Fact-Checking	classify	pct
factsopinions2025	Saju et al. (2025)	Fact-Checking	classify	accu
kaikaus2023financecorpora	Kaikaus et al. (2023)	Finance	classify	accu
bizfinbench2025	Lu et al. (2025)	Finance	pointwise	pct
trident2025	Hui et al. (2025)	Finance	classify	pct
chan2023chateval	Chan et al. (2023)	General NLG	pairwise	pct
chiang2023alt	Chiang & Lee (2023)	General NLG	pointwise	pct
dubois2023alpacaform	Dubois et al. (2023)	General NLG	pairwise	winr
hsu2023figurecaption	Hsu et al. (2023)	General NLG	pointwise	spea
kim2023prometheus	Kim et al. (2023)	General NLG	pointwise	pear
lai2023styletransfer	Lai et al. (2023)	General NLG	pointwise	spea
li2023autoj	Li et al. (2023)	General NLG	pairwise	pct
wang2023chatgptnlg	Wang et al. (2023)	General NLG	pointwise	spea
wang2023pandalm	Wang et al. (2023)	General NLG	pairwise	accu
xu2023instructscore	Xu et al. (2023)	General NLG	pointwise	pear
zhang2023widerdeeper	Zhang et al. (2023)	General NLG	pairwise	pct
zheng2023mtbench	Zheng et al. (2023)	General NLG	pairwise	pct
chiang2024arena	Chiang et al. (2024)	General NLG	pairwise	pct
doostmohammadi2024autoeval	Doostmohammadi et al. (2024)	General NLG	pointwise	pct
dubois2024lc	Dubois et al. (2024)	General NLG	pairwise	winr
hashemi2024llmrubric	Hashemi et al. (2024)	General NLG	pointwise	pear
huang2024nlexpl	Huang et al. (2024)	General NLG	pointwise	pear
kim2024prometheus2	Kim et al. (2024)	General NLG	pointwise	pear
li2024arenahard	Li et al. (2024)	General NLG	pairwise	pct
lin2024wildbench	Lin et al. (2024)	General NLG	pairwise	winr
verga2024poll	Verga et al. (2024)	General NLG	pairwise	pct
faggioli2023perspectives	Faggioli et al. (2023)	IR & Relevance	classify	kapp
alaofi2024fooledrelevant	Alaofi et al. (2024)	IR & Relevance	classify	pct
clarke2024cantreplace	Clarke & Dietz (2024)	IR & Relevance	classify	kapp
thomas2024relevance	Thomas et al. (2024)	IR & Relevance	classify	kapp
upadhyay2024largescale	Upadhyay et al. (2024)	IR & Relevance	classify	kapp
zhuang2024beyondyesno	Zhuang et al. (2024)	IR & Relevance	ranking	kend
arabzadeh2025benchrelevance	Arabzadeh & Clarke (2025)	IR & Relevance	classify	kapp
arabzadeh2025promptsens	Arabzadeh & Clarke (2025)	IR & Relevance	classify	kapp
frobe2025assessorsagree	Frobe et al. (2025)	IR & Relevance	classify	kapp
li2023coannotating	Li et al. (2023)	Information Extraction (NER/NLI)	classify	pct
humanllmcollab2024	Wang et al. (2024)	Information Extraction (NER/NLI)	classify	pct
kholodna2024activelearning	Kholodna et al. (2024)	Information Extraction (NER/NLI)	classify	f1
kim2024meganno	Kim et al. (2024)	Information Extraction (NER/NLI)	classify	f1
nasution2024chatgptlabel	Nasution & Onan (2024)	Information Extraction (NER/NLI)	classify	accu

Key	Study	Domain	Task	Metri
ner2024augment	Naraki et al. (2024)	Information Extraction (NER/NLI)	classify	f1
nli2024labeldist	Chen et al. (2024)	Information Extraction (NER/NLI)	classify	pear
ronningstad2024gptannotator	Ronningstad et al. (2024)	Information Extraction (NER/NLI)	classify	f1
qiu2025labelensemble	Qiu et al. (2025)	Information Extraction (NER/NLI)	classify	accu
zhou2023ifeval	Zhou et al. (2023)	Instruction Following	verify	accu
zeng2024llmbar	Zeng et al. (2024)	Instruction Following	pairwise	accu
ifcritic2025	Wen et al. (2025)	Instruction Following	classify	pct_
legaldocrec2025	Pradhan et al. (2025)	Legal	ranking	spea
lemaj2025	Enguehard et al. (2025)	Legal	pointwise	pct_
kocmi2023gemba	Kocmi & Federmann (2023)	Machine Translation	pointwise	kend
kocmi2023gembamqm	Kocmi & Federmann (2023)	Machine Translation	verify	pct_
lu2023erroranalysis	Lu et al. (2023)	Machine Translation	verify	kend
sun2023radimpressions	Sun et al. (2023)	Medicine & Clinical	pointwise	pct_
tang2023medevidence	Tang et al. (2023)	Medicine & Clinical	pointwise	pct_
brain2024radgpt	Mitsuyama et al. (2024)	Medicine & Clinical	classify	accu
brake2024clinicalnote	Brake & Schaaf (2024)	Medicine & Clinical	pointwise	pct_
wang2024clinicalevent	Wang et al. (2024)	Medicine & Clinical	classify	pct_
chung2025verifact	Chung et al. (2025)	Medicine & Clinical	verify	pct_
clinstruct2025	Brake et al. (2025)	Medicine & Clinical	pointwise	icc
croxford2025autoeval	Croxford et al. (2025)	Medicine & Clinical	pointwise	pct_
croxford2025clinsummjudge	Croxford et al. (2025)	Medicine & Clinical	pointwise	pct_
healthbench2025	Arora et al. (OpenAI) (2025)	Medicine & Clinical	pointwise	pct_
kim2025emulate	Kim & Kim (2025)	Medicine & Clinical	pointwise	pct_
llmevalmed2025	Zhang et al. (2025)	Medicine & Clinical	pointwise	pct_
szymanski2025limits	Szymanski et al. (2025)	Medicine & Clinical	pairwise	pct_
clindialog2026	Sun et al. (2026)	Medicine & Clinical	pointwise	pct_
frenchmedqa2026	Belmadani et al. (2026)	Medicine & Clinical	pointwise	kapp
medjudge2026scoping	Li et al. (2026)	Medicine & Clinical	review	nan
counselbench2025	Li et al. (2025)	Mental Health	pointwise	pct_
hua2025mhscoping	Hua et al. (2025)	Mental Health	review	nan
mhtrust2025	Badawi et al. (2025)	Mental Health	pointwise	icc
mhsupport2026	Badawi et al. (2026)	Mental Health	pointwise	icc
liu2023trustworthy	Liu et al. (2023)	Methodology & Bias	survey	nan
saito2023verbositybias	Saito et al. (2023)	Methodology & Bias	pairwise	pct_
wang2023shepherd	Wang et al. (2023)	Methodology & Bias	pointwise	pct_
zhu2023judgelm	Zhu et al. (2023)	Methodology & Bias	pairwise	pct_
bavaresco2024judgebench	Bavaresco et al. (2024)	Methodology & Bias	classify	kapp
chen2024humansorllms	Chen et al. (2024)	Methodology & Bias	pairwise	pct_
desmond2024evalullm	Desmond et al. (2024)	Methodology & Bias	pointwise	pct_
fan2024sedareval	Fan et al. (2024)	Methodology & Bias	pointwise	pct_
feuer2024style	Feuer et al. (2024)	Methodology & Bias	pairwise	pct_
gu2024survey	Gu et al. (2024)	Methodology & Bias	survey	nan
jung2024trustescalate	Jung et al. (2024)	Methodology & Bias	pairwise	pct_
justice2024prejudice	Ye et al. (2024)	Methodology & Bias	pairwise	pct_
ke2024critiquellm	Ke et al. (2024)	Methodology & Bias	pointwise	spea
koo2024cobbler	Koo et al. (2024)	Methodology & Bias	pairwise	wirr
li2024compsurvey	Li et al. (2024)	Methodology & Bias	survey	nan
li2024gen2judge	Li et al. (2024)	Methodology & Bias	survey	nan
li2024splitmerge	Li et al. (2024)	Methodology & Bias	pairwise	pct_
liu2024pairwisepref	Liu et al. (2024)	Methodology & Bias	pairwise	pct_
pan2024hcdjudge	Pan et al. (2024)	Methodology & Bias	pointwise	nan
panickssery2024selfpref	Panickssery et al. (2024)	Methodology & Bias	pairwise	pct_
raina2024robust	Raina et al. (2024)	Methodology & Bias	pointwise	pct_
shankar2024validators	Shankar et al. (2024)	Methodology & Bias	pointwise	pct_
stureborg2024inconsistent	Stureborg et al. (2024)	Methodology & Bias	pointwise	pct_
thakur2024judgingjudges	Thakur et al. (2024)	Methodology & Bias	pairwise	pct_
wataoka2024selfpref	Wataoka et al. (2024)	Methodology & Bias	pairwise	pct_

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zhu2024nationalitybias	Zhu et al. (2024)	Methodology & Bias	pointwise	pct_
calderon2025alttest	Calderon et al. (2025)	Methodology & Bias	classify	pct_
gao2025nlgsurvey	Gao et al. (2025)	Methodology & Bias	survey	nan
gao2025reranking	Gao et al. (2025)	Methodology & Bias	ranking	kend
hu2025trainjudge	Hu et al. (2025)	Methodology & Bias	pointwise	pct_
huang2025empiricaljudge	Huang et al. (2025)	Methodology & Bias	pointwise	pct_
judgesverdict2025	Han et al. (2025)	Methodology & Bias	pointwise	kapp
li2025prefleakage	Li et al. (2025)	Methodology & Bias	pairwise	pct_
murugadoss2025evaltheeval	Murugadoss et al. (2025)	Methodology & Bias	pointwise	pct_
voutsas2025biaseddesign	Voutsas et al. (2025)	Methodology & Bias	classify	pct_
yamauchi2025designchoices	Yamauchi et al. (2025)	Methodology & Bias	pointwise	pct_
imrit2026iaa	James et al. (2026)	Methodology & Bias	review	kripp
temperature2026	Li et al. (2026)	Methodology & Bias	pointwise	pct_
khondaker2023gptaraeval	Khondaker et al. (2023)	Multilingual	classify	f1
mmeval2024	Son et al. (2024)	Multilingual	pairwise	accu
multiljudge2025	Fu et al. (2025)	Multilingual	pointwise	pct_
reliablemulti2026	Zubiaga et al. (2026)	Multilingual	pointwise	pct_
chen2024mllmjudge	Chen et al. (2024)	Multimodal (Vision-Language)	pairwise	pct_
duan2024uimockup	Duan et al. (2024)	Multimodal (Vision-Language)	pointwise	pct_
lee2024promvision	Lee et al. (2024)	Multimodal (Vision-Language)	pointwise	pear
lu2024wildvision	Lu et al. (2024)	Multimodal (Vision-Language)	pairwise	pct_
yasunaga2025mmrewardbench	Yasunaga et al. (2025)	Multimodal (Vision-Language)	pairwise	accu
chatgptreviewers2024	Ho et al. (2024)	Peer Review & Science	pointwise	pct_
liang2024feedback	Liang et al. (2024)	Peer Review & Science	pointwise	pct_
murad2024quality	Tarakji et al. (2024)	Peer Review & Science	classify	pct_
prismascreen2024	Guo et al. (2024)	Peer Review & Science	classify	kapp
rose2025rob	Rose et al. (2025)	Peer Review & Science	classify	kapp
shopovski2025peerreview	Shopovski et al. (2025)	Peer Review & Science	pointwise	pct_
guo2023howclose	Guo et al. (2023)	QA & Factuality	pointwise	pct_
min2023factscore	Min et al. (2023)	QA & Factuality	verify	pct_
li2024pedants	Li et al. (2024)	QA & Factuality	verify	pct_
madaan2024refverdict	Reference-Guided Verdict authors (2024)	QA & Factuality	pointwise	pct_
wei2024safe	Wei et al. (2024)	QA & Factuality	verify	pct_
leas2023contentanalysis	Leas et al. (2023)	Qualitative Coding	classify	pct_
meng2024chalet	Meng et al. (2024)	Qualitative Coding	classify	pct_
qualigpt2024	Zhang et al. (2024)	Qualitative Coding	classify	pct_
mehta2025qualgenai	Mehta et al. (2025)	Qualitative Coding	classify	pct_
qualcoding2025jla	Xiao et al. (2025)	Qualitative Coding	classify	fleiss
thematic2025charity	Wen et al. (2025)	Qualitative Coding	classify	pct_
es2023ragas	Es et al. (2023)	RAG	pointwise	pct_
saadfalcon2023ares	Saad-Falcon et al. (2023)	RAG	classify	accu
jin2024ragreward	Jin et al. (2024)	RAG	pairwise	accu
raju2024calc2adj	Raju et al. (2024)	Reasoning & Math	verify	accu
tan2024judgebench	Tan et al. (2024)	Reasoning & Math	pairwise	accu
mathrobust2026	Yosef et al. (2026)	Reasoning & Math	verify	accu
lee2023rlaif	Lee et al. (2023)	Reward Modeling	pairwise	pct_
lambert2024rewardbench	Lambert et al. (2024)	Reward Modeling	pairwise	accu
wu2024metarewarding	Wu et al. (2024)	Reward Modeling	pairwise	pct_
yuan2024selfrewarding	Yuan et al. (2024)	Reward Modeling	pairwise	pct_
ifreward2026	Wen et al. (2026)	Reward Modeling	ranking	kend
belal2023sentiment	Belal et al. (2023)	Sentiment & Emotion	classify	accu
koptyra2023clarinemo	Koptyra et al. (2023)	Sentiment & Emotion	classify	pct_
zhang2023sentiment	Zhang et al. (2023)	Sentiment & Emotion	classify	accu
niu2024text2emotion	Niu et al. (2024)	Sentiment & Emotion	classify	pct_
aldeen2023chatgptvshuman	Aldeen et al. (2023)	Social-Science Annotation	classify	accu
cegin2023paraphrases	Cegin et al. (2023)	Social-Science Annotation	classify	pct_
gilardi2023chatgpt	Gilardi et al. (2023)	Social-Science Annotation	classify	accu

Key	Study	Domain	Task	Metri
ollion2023mindhype	Ollion et al. (2023)	Social-Science Annotation	classify	pct_
ostyakova2023crowdexperts	Ostyakova et al. (2023)	Social-Science Annotation	classify	pct_
pangakis2023validation	Pangakis et al. (2023)	Social-Science Annotation	classify	f1
reiss2023testing	Reiss (2023)	Social-Science Annotation	classify	pct_
wu2023llmworkers	Wu et al. (2023)	Social-Science Annotation	classify	pct_
zhu2023reproduce	Zhu et al. (2023)	Social-Science Annotation	classify	pct_
li2024stance	Li & Conrad (2024)	Social-Science Annotation	classify	pct_
pangakis2024keephumans	Pangakis & Wolken (2024)	Social-Science Annotation	classify	f1
ziems2024css	Ziems et al. (2024)	Social-Science Annotation	classify	accu
ahmad2025moralmachine	Ahmad & Takemoto (2025)	Social-Science Annotation	classify	pct_
horych2025promisespitfalls	Horych et al. (2025)	Social-Science Annotation	classify	f1
tornberg2025political	Törnberg (2025)	Social-Science Annotation	classify	accu
gao2023humanlike	Gao et al. (2023)	Summarization	pointwise	spea
liu2023geval	Liu et al. (2023)	Summarization	pointwise	spea
luo2023factinconsistency	Luo et al. (2023)	Summarization	verify	pct_
shen2023notyet	Shen et al. (2023)	Summarization	pointwise	spea
lee2024unisumeval	Lee et al. (2024)	Summarization	pointwise	pct_
song2024finesure	Song et al. (2024)	Summarization	verify	pct_
tang2024tofueval	Tang et al. (2024)	Summarization	verify	pct_
zhang2024newssumm	Zhang et al. (2024)	Summarization	pointwise	pct_
wu2023hpsv2	Wu et al. (2023)	Text-to-Image	pointwise	accu
lin2024vqascore	Lin et al. (2024)	Text-to-Image	pointwise	pear

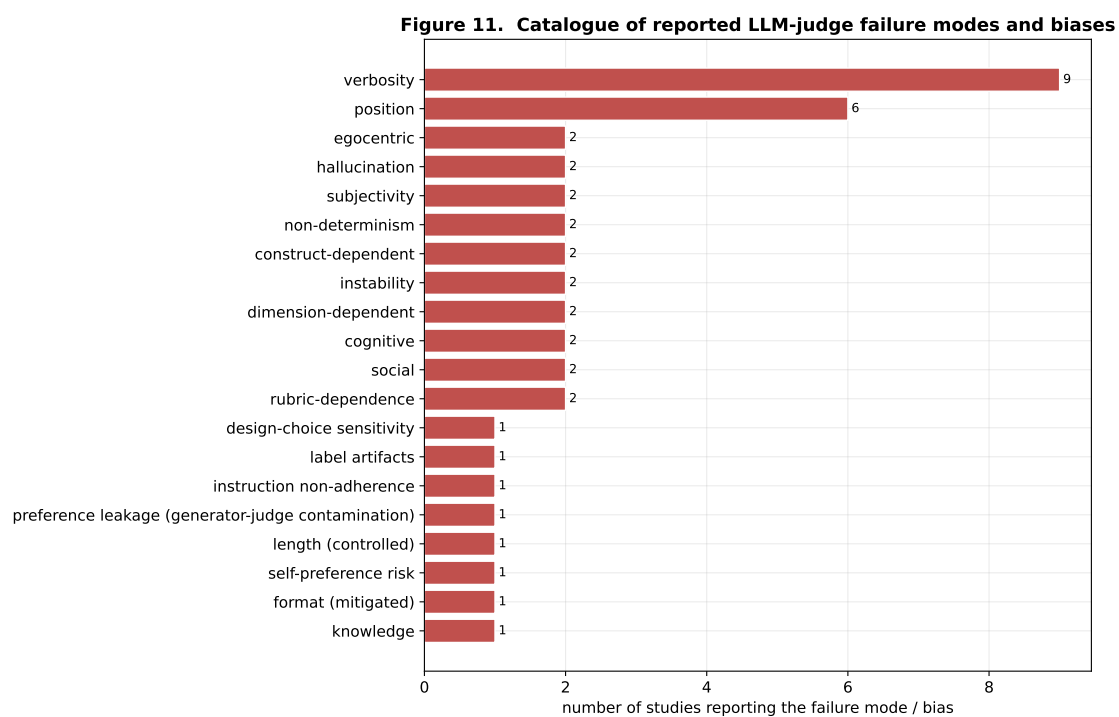


Figure 10: Reported failure modes / biases (study counts).

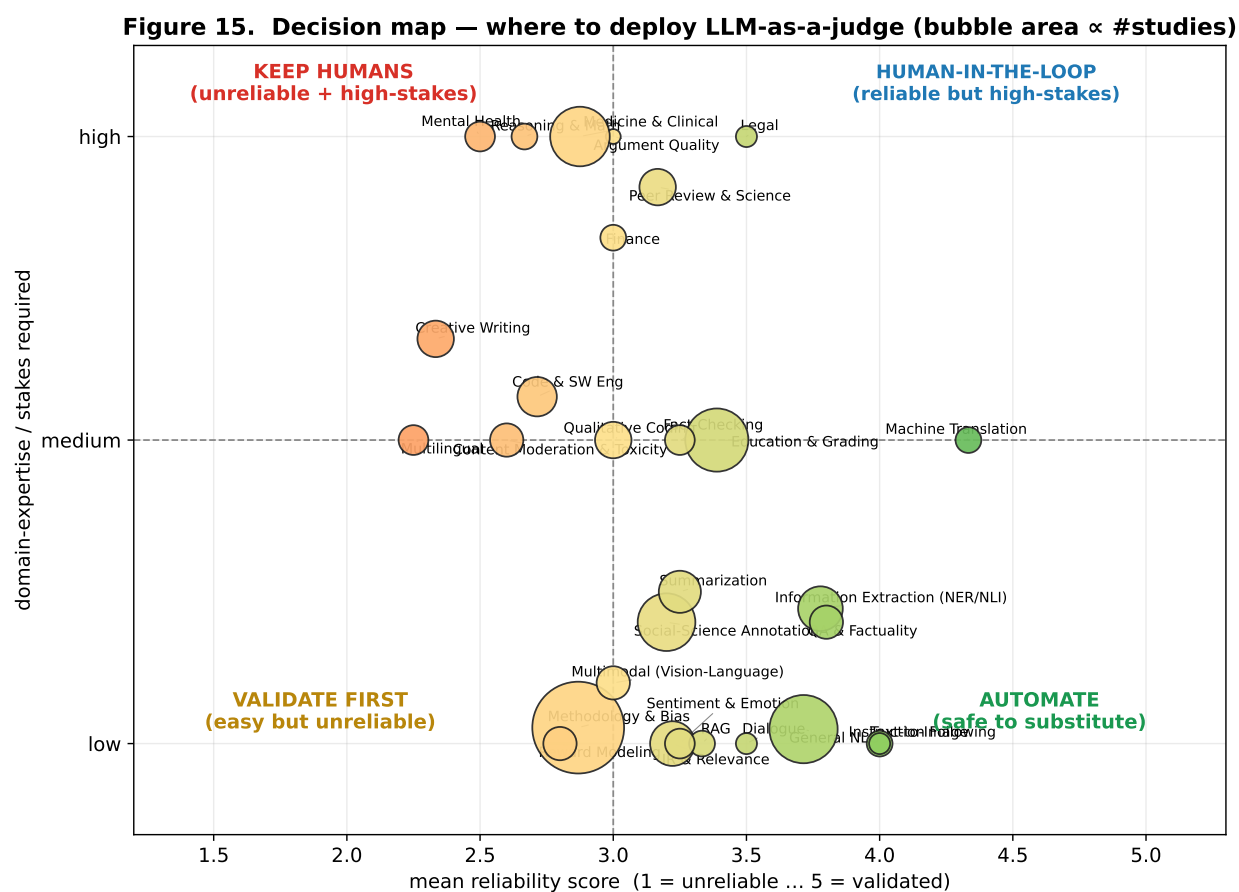


Figure 11: Deployment decision map (reliability $\times$ stakes; bubble area $\propto$ studies).